

Online Appendix:

Does Money Buy National Happiness? It Depends on Who You Ask

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Section numbers refer to the main paper.

Table A1: Correlation between national well-being indicators (WVS country-years).

	Happy	Very Happy	Very Unhappy	V. Happy – V. Unhappy	Happiness (mean)	Satisfaction (mean)	Satisfied	Happy + Satisfied
Happy	1.00	0.60	–0.73	0.70	0.89	0.73	0.72	0.90
Very Happy	0.60	1.00	–0.37	0.98	0.89	0.60	0.51	0.59
Very Unhappy	–0.73	–0.37	1.00	–0.54	–0.67	–0.58	–0.57	–0.69
V. Happy – V. Unhappy	0.70	0.98	–0.54	1.00	0.94	0.66	0.58	0.67
Happiness (mean)	0.89	0.89	–0.67	0.94	1.00	0.75	0.69	0.83
Satisfaction (mean)	0.73	0.60	–0.58	0.66	0.75	1.00	0.96	0.93
Satisfied	0.72	0.51	–0.57	0.58	0.69	0.96	1.00	0.95
Happy + Satisfied	0.90	0.59	–0.69	0.67	0.83	0.93	0.95	1.00

(Back to Section [2.1.](#))

Table A2: Slope of national well-being indicators with respect to $\log_{10}$ GDP per capita (PPP).

Data	Indicator	Slope	s.e.	p-value	R^2	Obs.
WVS	Happy	0.109	0.019	0.000	0.144	340
WVS	Very Happy	0.002	0.025	0.948	0.000	340
WVS	Very Unhappy	−0.022	0.004	0.000	0.072	340
WVS	V. Happy – V. Unhappy	0.023	0.026	0.379	0.004	340
WVS	Happiness (mean)	0.264	0.083	0.002	0.044	340
WVS	Satisfaction (mean)	1.143	0.142	0.000	0.233	340
WVS	Satisfied	0.236	0.024	0.000	0.316	340
WVS	Happy + Satisfied	0.173	0.020	0.000	0.275	338
Gallup	Satisfied	0.372	0.017	0.000	0.655	2344
Gallup	Very satisfied (8–10)	0.218	0.018	0.000	0.442	2344
Gallup	Extremely satisfied (9–10)	0.065	0.009	0.000	0.178	2344
Gallup	Completely satisfied (10)	0.012	0.005	0.008	0.015	2344
Gallup	Unsatisfied (0/1–4)	−0.304	0.012	0.000	0.639	2344
Gallup	Dissatisfied (0/1–2)	−0.135	0.009	0.000	0.432	2344
Gallup	Satisfaction (mean)	1.779	0.083	0.000	0.611	2344

Note: OLS across country-years; standard errors clustered by country. Gallup: 2006–2023. (Back to Section 3.2.)

Table A3: Out-of-sample predictive capacity: leave-one-country-out cross-validated R^2 .

Data	Indicator	Obs.	log GDP PPP	Best income variable	Region	log GDP PPP + Region
WVS, all waves	Happy	340	0.12	0.18	0.26	0.32
WVS, all waves	Very Happy	340	−0.03	−0.04	0.27	0.27
WVS, all waves	Very Unhappy	340	0.06	0.10	0.10	0.11
WVS, all waves	V. Happy – V. Unhappy	340	−0.02	−0.02	0.27	0.26
WVS, all waves	Happiness (mean)	340	0.02	0.05	0.29	0.30
WVS, all waves	Satisfaction (mean)	340	0.21	0.23	0.37	0.46
WVS, all waves	Satisfied	340	0.30	0.34	0.36	0.50
WVS, all waves	Happy + Satisfied	338	0.26	0.31	0.35	0.47
WVS, last obs. per country	Happy	115	0.05	0.03	0.11	0.12
WVS, last obs. per country	Very Happy	115	0.01	−0.05	0.17	0.16
WVS, last obs. per country	Very Unhappy	115	0.04	0.06	0.01	0.02
WVS, last obs. per country	V. Happy – V. Unhappy	115	−0.01	−0.08	0.16	0.15
WVS, last obs. per country	Happiness (mean)	115	−0.04	−0.09	0.12	0.10
WVS, last obs. per country	Satisfaction (mean)	115	0.18	0.19	0.30	0.33
WVS, last obs. per country	Satisfied	115	0.29	0.31	0.31	0.38
WVS, last obs. per country	Happy + Satisfied	115	0.24	0.25	0.28	0.34
Gallup, all years (2006–23)	Satisfaction (mean)	2344	0.61	0.66	0.49	0.68
Gallup, all years (2006–23)	Satisfied	2344	0.65	0.72	0.52	0.72
Gallup, all years (2006–23)	Unsatisfied (0/1–4)	2344	0.63	0.65	0.48	0.68
Gallup, last obs. per country	Satisfaction (mean)	167	0.60	0.58	0.45	0.64
Gallup, last obs. per country	Satisfied	167	0.65	0.68	0.52	0.70
Gallup, last obs. per country	Unsatisfied (0/1–4)	167	0.65	0.60	0.48	0.67
WHR, 2023–25 average	Satisfaction (mean)	147	0.62	0.60	0.52	0.69

Note: Each country's observations are predicted with a model estimated on all other countries. Cross-validated $R^2 = 1 - \sum(y - \hat{y}_{\text{out-of-sample}})^2 / \sum(y - \bar{y})^2$; it can be negative when a model predicts worse than the overall mean. In each row, the highest cross-validated R^2 among log GDP PPP, the best income variable and region is in bold. (Back to Section 3.3.)

Table A4: Robustness: income vs. region across specifications and datasets.

Specification	Obs.	Countries	R^2 income		R^2 region	Share of explained variance due to income		Region better (share of cases)
			log PPP	best		log PPP	best	
WVS, all waves	340	112	0.14	0.20	0.33	0.23	0.33	0.27
WVS, waves 1–2 (1981–91)	26	21	0.07	0.43	0.58	0.09	0.40	0.10
WVS, wave 3 (1995–99)	55	55	0.16	0.33	0.65	0.14	0.28	0.18
WVS, wave 4 (1999–2004)	40	40	0.19	0.31	0.33	0.27	0.47	0.47
WVS, wave 5 (2004–09)	58	58	0.21	0.35	0.49	0.22	0.38	0.23
WVS, wave 6 (2010–16)	60	60	0.08	0.22	0.27	0.14	0.41	0.19
WVS, wave 7 (2017–23)	102	92	0.09	0.17	0.28	0.21	0.35	0.27
WVS, wave 7 w/o 2020–21	80	72	0.08	0.19	0.29	0.19	0.38	0.27
WVS, population-weighted	340	112	0.10	0.19	0.21	0.25	0.44	0.32
WVS, last obs. per country	115	112	0.13	0.19	0.25	0.26	0.39	0.30
WVS, no GDP imputation	321	108	0.13	0.20	0.32	0.22	0.33	0.27
WVS, GDP PPP 2017 \$ (old)	340	112	0.14	0.20	0.33	0.24	0.33	0.28
WVS, without EVS surveys	304	108	0.12	0.19	0.37	0.18	0.28	0.22
Gallup, all years (2006–23)	2344	167	0.42	0.48	0.42	0.46	0.51	0.49
Gallup, last obs. per country	167	167	0.42	0.44	0.41	0.44	0.48	0.45
Gallup, last obs., WVS countries	109	109	0.44	0.47	0.43	0.46	0.49	0.47
Gallup, last obs., 6 regions	167	167	0.42	0.44	0.42	0.43	0.46	0.44
Gallup, last obs., WB regions	167	167	0.42	0.44	0.38	0.47	0.50	0.48
WHR, 2023–25 average	147	147	0.63	0.63	0.55	0.56	0.56	0.56

Note: Averages over well-being indicators (8 for the WVS; 7 satisfaction indicators for Gallup; mean ladder for the WHR). “Best” refers to the income measure with the highest average R^2 in the main WVS specification (Cluster k=7 nominal). “Region better” is the share of indicator $\times$ income-measure combinations for which $s < 0.5$; values above 0.5 are in bold. Alternative region classifications and exclusions of regions: see Table 3. (Back to Section 3.3.)

Table A5: Other country-level correlates of national well-being: Shapley decomposition of R^2 (WVS).

Indicator	Other variable	Obs.	R^2 other alone	Shapley decomposition of R^2			
				Income	Region	Other	Total
Satisfaction (mean)	Freedom of choice	334	0.63	0.11	0.21	0.42	0.74
Satisfaction (mean)	Importance of God	329	0.01	0.17	0.33	0.02	0.51
Satisfaction (mean)	Tolerance (homosexuality)	326	0.24	0.12	0.30	0.10	0.52
Satisfaction (mean)	Importance of democracy	216	0.08	0.13	0.32	0.02	0.47
Satisfaction (mean)	GDP growth (5-year)	334	0.02	0.16	0.34	0.02	0.52
Happiness (mean)	Freedom of choice	333	0.39	0.02	0.22	0.27	0.51
Happiness (mean)	Importance of God	330	0.01	0.05	0.31	0.03	0.39
Happiness (mean)	Tolerance (homosexuality)	325	0.09	0.02	0.31	0.07	0.40
Happiness (mean)	Importance of democracy	216	0.03	0.02	0.28	−0.03	0.27
Happiness (mean)	GDP growth (5-year)	334	0.01	0.03	0.31	0.00	0.34

Note: Country-year averages of WVS variables: perceived freedom of choice and control (1–10), importance of God in one’s life (1–10), justifiability of homosexuality (1–10), importance of living in a democracy (1–10, waves 5–7), and average annual growth of real GDP per capita over the previous five years. Shapley decomposition of the R^2 of the regression on log GDP per capita, region dummies and the other variable. In each row, the largest contribution among income, region and the other variable is in bold. (Back to Section 3.6.)

Table A6: Within-country relation between national well-being and $\log_{10}$ GDP per capita (country fixed effects, WVS).

Indicator	Slope	s.e.	<i>p</i> -value	Within R^2	Obs.
Happy	0.286	0.043	0.000	0.281	312
Very Happy	0.207	0.048	0.000	0.146	312
Very Unhappy	−0.039	0.010	0.000	0.044	312
V. Happy – V. Unhappy	0.246	0.054	0.000	0.160	312
Happiness (mean)	1.063	0.180	0.000	0.245	312
Satisfaction (mean)	2.313	0.420	0.000	0.344	312
Satisfied	0.440	0.076	0.000	0.347	312
Happy + Satisfied	0.366	0.058	0.000	0.389	310

Note: Countries surveyed at least twice. Standard errors clustered by country. On average over the eight indicators, the within-country R^2 is 0.24. (Back to Section 3.6.)

Table A7: Countries most often ranked happiest (WVS, 8 indicators $\times$ 8 samples: each wave and all waves combined).

Country	Occurrences as happiest	Region
Switzerland	10	Western
Mexico	9	Latin America
Kyrgyzstan	6	Asia
Australia	5	Western
Netherlands	4	Western
Nigeria	4	Africa
Puerto Rico	4	Latin America
Colombia	3	Latin America
Finland	3	Western
Vietnam	3	Asia

Note: Looking at all waves combined, the happiest country-year is: Kyrgyzstan–2020 (3 indicators: Very Happy, V. Happy – V. Unhappy, Happiness (mean)); Finland–1981 (Satisfied); Malaysia–2012 (Very Unhappy); Mexico–2012 (Satisfaction (mean)); Qatar–2010 (Happy); Vietnam–2020 (Happy + Satisfied). For *Very Unhappy*, the happiest country-year is the one with the lowest share. (Back to Section 3.6.)

Table A8: Effects of question wording and scale on life evaluation, Fabre (2025) survey experiment.

Dependent Variables: Model:	Answer		Answer (0–10 equiv.)	Satisfied (6+)	Answer (0–10 equiv.)	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Ladder wording (Gallup)	-0.451*** (0.046)	-0.455*** (0.067)	-0.456*** (0.067)	-0.088*** (0.013)	-0.464*** (0.123)	-0.494** (0.172)
Scale 1–10	0.137** (0.046)	0.132 (0.068)	-0.260*** (0.072)	-0.008 (0.013)	-0.276*** (0.046)	-0.540** (0.182)
Ladder wording (Gallup) × Scale 1–10		0.008 (0.092)	-0.041 (0.097)	0.028 (0.019)		0.055 (0.247)
Income quartile (1–4)					0.689*** (0.031)	0.635*** (0.043)
Income quartile (1–4) × Ladder wording (Gallup)					-0.004 (0.042)	0.009 (0.058)
Income quartile (1–4) × Scale 1–10						0.106 (0.061)
Income quartile (1–4) × Ladder wording (Gallup) × Scale 1–10						-0.022 (0.083)
<i>Fixed-effects</i>						
Country	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	11,000	11,000	11,000	11,000	11,000	11,000
R ²	0.05681	0.05681	0.05942	0.04208	0.15976	0.16024

Heteroskedasticity-robust standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05*

Note: OLS with country fixed effects, survey weights (normalized within country) and heteroskedasticity-robust standard errors in parentheses. *Ladder wording:* Cantril ladder (vs. WVS life satisfaction). *Scale 1–10:* answers from 1 to 10 (vs. 0 to 10). In columns 3, 5 and 6, 1–10 answers are linearly stretched to 0–10. Columns 5 and 6 interact the treatments with the respondent’s income quartile within their country; in column 6, the interaction between income and wording is 0.009 ($p = .883$) and the triple interaction is -0.023 ($p = .786$). * $p < .05$, ** $p < .01$, *** $p < .001$. (Back to Section 4.3.)

Table A9: Decomposition of the WVS–Gallup gap by country.

Country	Year WVS/EVS (Gallup)	Survey: mode	Gallup (1)	WVS/EVS (2)	Gap D (2)–(1)	Ladder 0–10 (3)	Satisf. 1–10 (4)	Question effect Q (4)–(3)	Residual R $D - Q$
Poland	2017	EVS: CAPI	6.28	7.56	1.28	5.53	6.37	0.84	0.44
Spain	2017	EVS: CAPI	6.47	7.49	1.02	5.61	6.99	1.38	-0.36
Japan	2019	WVS: Mail	5.94	6.76	0.82	5.33	5.00	-0.33	1.15
Italy	2018	EVS: CAPI	6.69	7.28	0.59	5.99	6.59	0.60	-0.01
Germany	2018	WVS: CAPI	7.15	7.74	0.58	6.01	6.84	0.83	-0.25
United Kingdom	2022	WVS: CAPI/Web/Mail	6.76	7.32	0.56	5.67	6.77	1.10	-0.55
France	2018	EVS: CAPI	6.79	7.26	0.47	5.96	6.68	0.72	-0.25
Switzerland	2017	EVS: Web/Mail/CAPI	7.58	8.02	0.43	6.85	7.31	0.46	-0.03
Saudi Arabia	2003 (2006)	WVS: Face-to-face	7.06	7.28	0.23	6.31	7.10	0.79	-0.57
United States	2017	WVS: Web/Phone	7.23	7.22	-0.01	6.03	6.74	0.70	-0.71

Note: Mean answers on native scales (Gallup and ladder: 0–10; WVS/EVS and satisfaction: 1–10); columns (3) and (4): Fabre (2025). The Gallup year is given in parentheses when it differs from the WVS year. Survey: mode: source of the WVS/EVS data and main interview mode(s) (CAPI: computer-assisted personal interviews). (Back to Section 4.3.)

Table A10: Decomposition of the WVS–Gallup gap: summary.

Statistic	Estimate	95% bootstrap CI
Mean gap D (WVS – Gallup)	0.60	[0.55; 0.64]
Mean question effect Q (satisfaction 1–10 – ladder 0–10)	0.71	[0.56; 0.84]
Mean residual $R = D - Q$ (samples, mode, context)	−0.11	[−0.26; 0.04]
Share of mean gap due to question ($\bar{Q}/\bar{D}$)	1.19	[0.94; 1.45]
Share of gross level difference due to question ($ \bar{Q} /(\bar{Q} + \bar{R})$)	0.86	[0.76; 0.99]
Share of cross-country variance of D due to Q ($\text{cov}(D, Q)/\text{var}(D)$)	0.11	[−0.33; 0.55]
Correlation between D and Q	0.09	[−0.26; 0.41]
Mean absolute gap $ D $	0.60	[0.56; 0.65]
Mean absolute residual $ R $	0.43	[0.33; 0.60]
Mean gap, share satisfied (6+)	0.07	[0.06; 0.08]
Mean question effect, share satisfied (6+)	0.11	[0.08; 0.14]
Variance share due to Q , share satisfied	0.04	[−0.33; 0.45]

Note: 10 countries. 95% confidence intervals from 1,000 bootstrap replications resampling respondents within country $\times$ variant (Fabre) and within country (WVS). For Gallup, only the distribution of answers is available: in each country, the numbers of respondents giving each answer (0 to 10) are redrawn from a multinomial distribution with the observed shares and sample size, which is equivalent to resampling respondents. (Back to Section 4.3.)

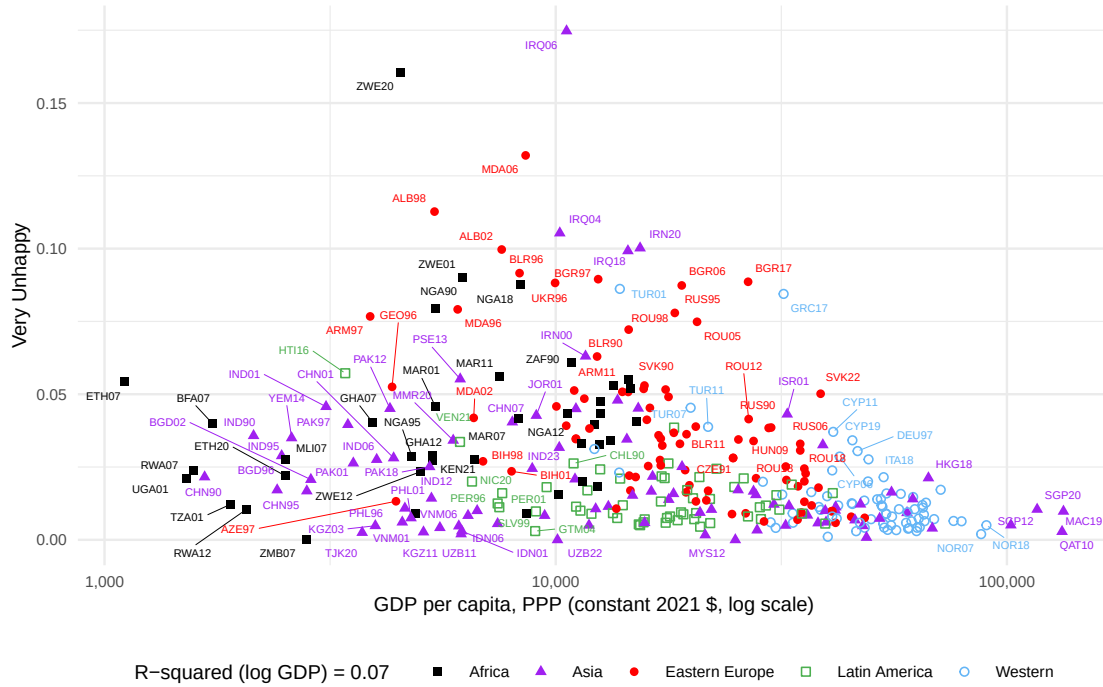
Table A11: Representativeness of WVS/EVS samples: sample shares relative to population benchmarks.

Measure	Surveys	Median				Correlation with log GDP
		All	GDP < 10k	10–30k	> 30k	
Tertiary education, 25+ (ratio sample/benchmark)	295	1.18	1.98	1.18	1.05	−0.40
Urban (ratio sample/benchmark)	76	1.44	2.19	1.45	1.22	−0.69
Aged 65+ (ratio sample/benchmark)	335	0.90	0.64	0.88	1.01	0.33
Women among married respondents − 0.5	337	0.01	−0.00	0.01	0.01	0.19

Note: One observation per WVS or EVS survey. Ratios: weighted sample share divided by the population share, medians across surveys (a ratio above 1 indicates over-representation). Benchmarks (World Bank WDI): share of the population aged 25+ with at least short-cycle tertiary education (UNESCO), or with complete or incomplete tertiary education (Barro & Lee, 2013) before 2011; urban population share (national definitions; sample: urban and peri-urban respondents, waves 4, 5 and 7 only); share aged 65+ among the population aged 15+ (sample: 65+ among respondents, usually 18+). Women among married: unweighted share of women among married or cohabiting respondents minus 0.5. Correlations with log GDP per capita (PPP) of the log ratios (or of the deviation). (Back to Section 4.4.)

Scatter plot showing GDP per capita, PPP (constant 2021 \$, log scale) on the X-axis (ranging from 1,000 to 100,000) versus Happy (R-squared (log GDP) = 0.14) on the Y-axis (ranging from 0.4 to 1.0). The plot displays data points for various countries, categorized by region: Africa (black squares), Asia (purple triangles), Eastern Europe (red circles), Latin America (green squares), and Western (blue circles). Numerous countries are labeled with codes (e.g., TZA01, RWA12, ETH07, etc.). The plot illustrates a positive correlation between GDP and happiness, with many countries clustered at lower GDP and happiness levels, and a few outliers at higher GDP and happiness levels.

Figure A3: *Very Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.



Note: Egypt 2013 (42% of very unhappy people) is not shown, to keep the y -axis readable; the R^2 in the legend includes it.

Figure A4: *V. Happy – V. Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

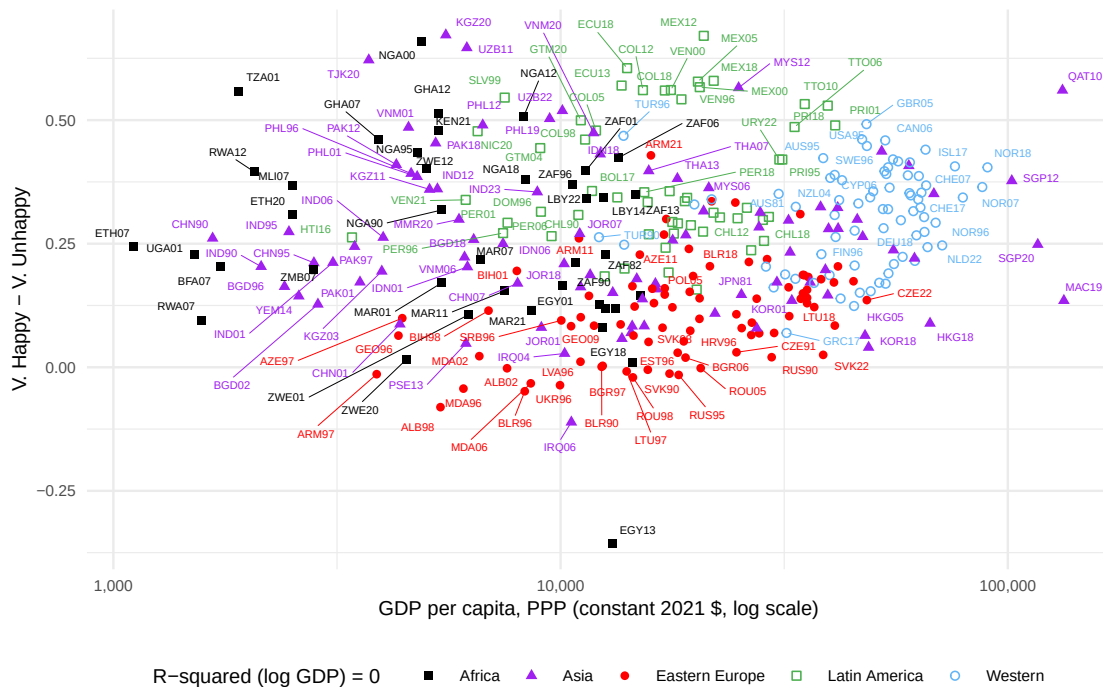


Figure A5: Satisfied vs. GDP per capita (PPP), WVS/EVS 1981–2023.

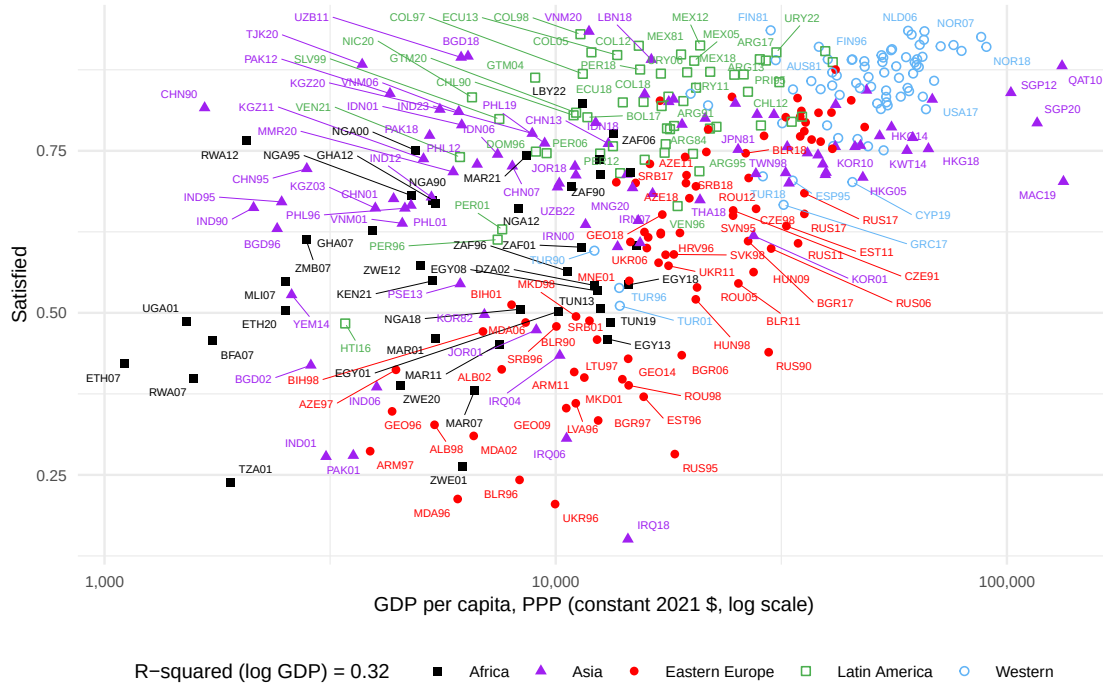


Figure A6: Happy + Satisfied vs. GDP per capita (PPP), WVS/EVS 1981–2023.

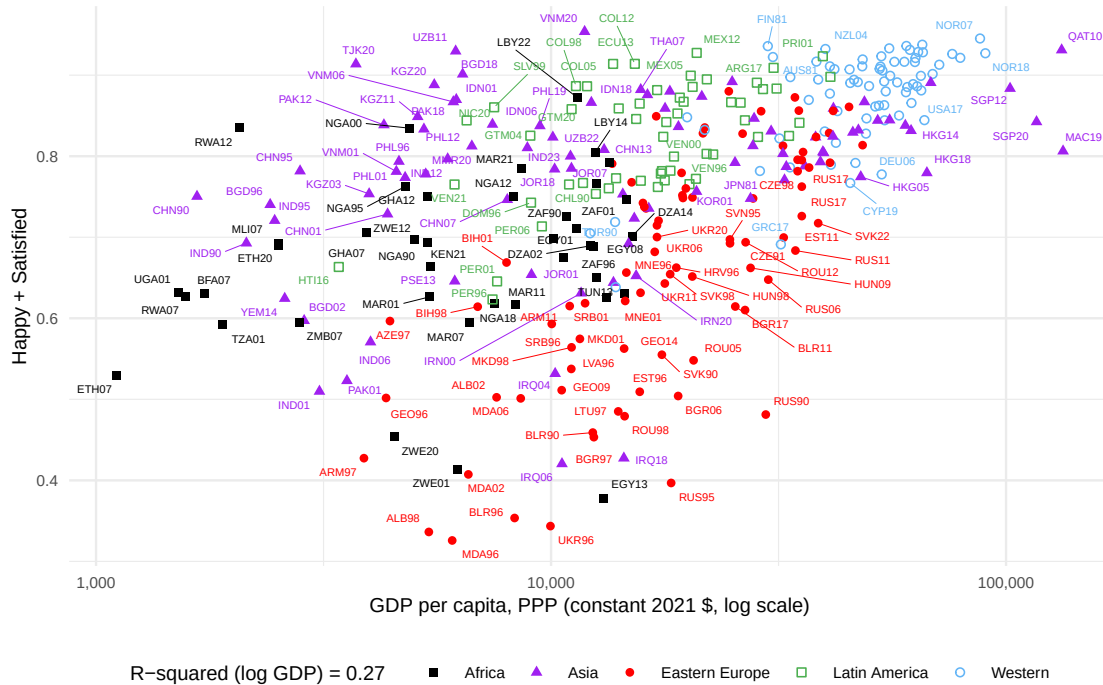
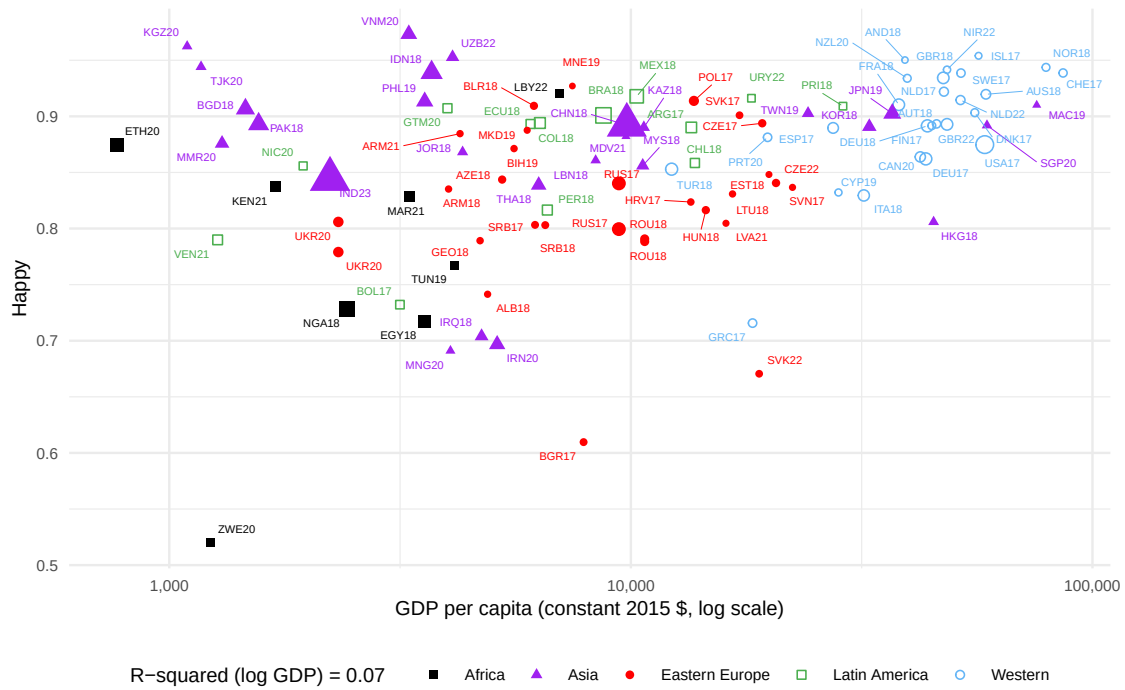


Figure A7: *Happy* vs. GDP per capita (nominal), WVS/EVS wave 7 (2017–2023), weighted by population.



Note: Point size and regression weights proportional to population.

(Back to Section 3.2.)

References

R. J. Barro & J. W. Lee. A new data set of educational attainment in the world, 1950–2010.
Journal of Development Economics, 2013. [6](#)